

Task 04 - Question Answers

C# Fundamentals - Enums & String vs StringBuilder

2. What's the Enum data type, when is it used? And name three common built-in enums used frequently.

An **enum** (enumeration) is a value type in C# that lets you define a named set of related constants under a single type, making code more readable and less error-prone than using plain numbers or strings to represent a fixed set of options. For example, instead of representing a day of the week as an integer (0, 1, 2...), you can define an enum with named members like Monday, Tuesday, Wednesday, so the code reads clearly and the compiler can catch invalid values.

Enums are used whenever a variable should only ever hold one of a small, fixed, known set of related values - such as a status, a category, a direction, or a mode - rather than an open-ended range of numbers or strings. This improves type safety (the compiler rejects values outside the defined set), readability, and maintainability, since the valid options are documented directly in the type itself.

Three common built-in enums used frequently in .NET are:

- **DayOfWeek** - represents the day of the week (Sunday, Monday, ... Saturday), commonly used with DateTime.
- **ConsoleColor** - represents the available foreground/background colors for console output (e.g., Console.ForegroundColor = ConsoleColor.Red).
- **DateTimeKind** - indicates whether a DateTime value represents a local time, UTC time, or an unspecified time (Local, Utc, Unspecified).

3. What are the scenarios to use string Vs StringBuilder?

The choice between string and StringBuilder comes down to how much and how often the text is going to change:

Use string when:

- The value is set once and rarely or never modified afterward (e.g., a name, a fixed message, a constant label).
- Only a small number of concatenations or modifications are needed (a handful of + operations or string interpolations have negligible overhead).
- The text needs to be compared, stored as a dictionary key, or passed around as an immutable value, since immutability guarantees it won't change unexpectedly and makes it inherently thread-safe.
- Readability matters more than raw performance, since string interpolation ("...") tends to be the clearest way to build simple, one-off strings.

Use StringBuilder when:

- Building or modifying text repeatedly, especially inside a loop (e.g., appending thousands of log

lines, building a large report, or generating HTML/CSV content dynamically).

- Performing many sequential `Append()`, `Insert()`, `Replace()`, or `Remove()` operations, since each of these on a plain string would otherwise allocate a brand-new string object every time.
- The final size of the text is large or not known in advance, and performance/memory efficiency matters more than the convenience of plain string syntax.

In short: `string` is the right default for simple, mostly-static text, while `StringBuilder` is the right tool whenever text is assembled or changed frequently and performance at scale matters.